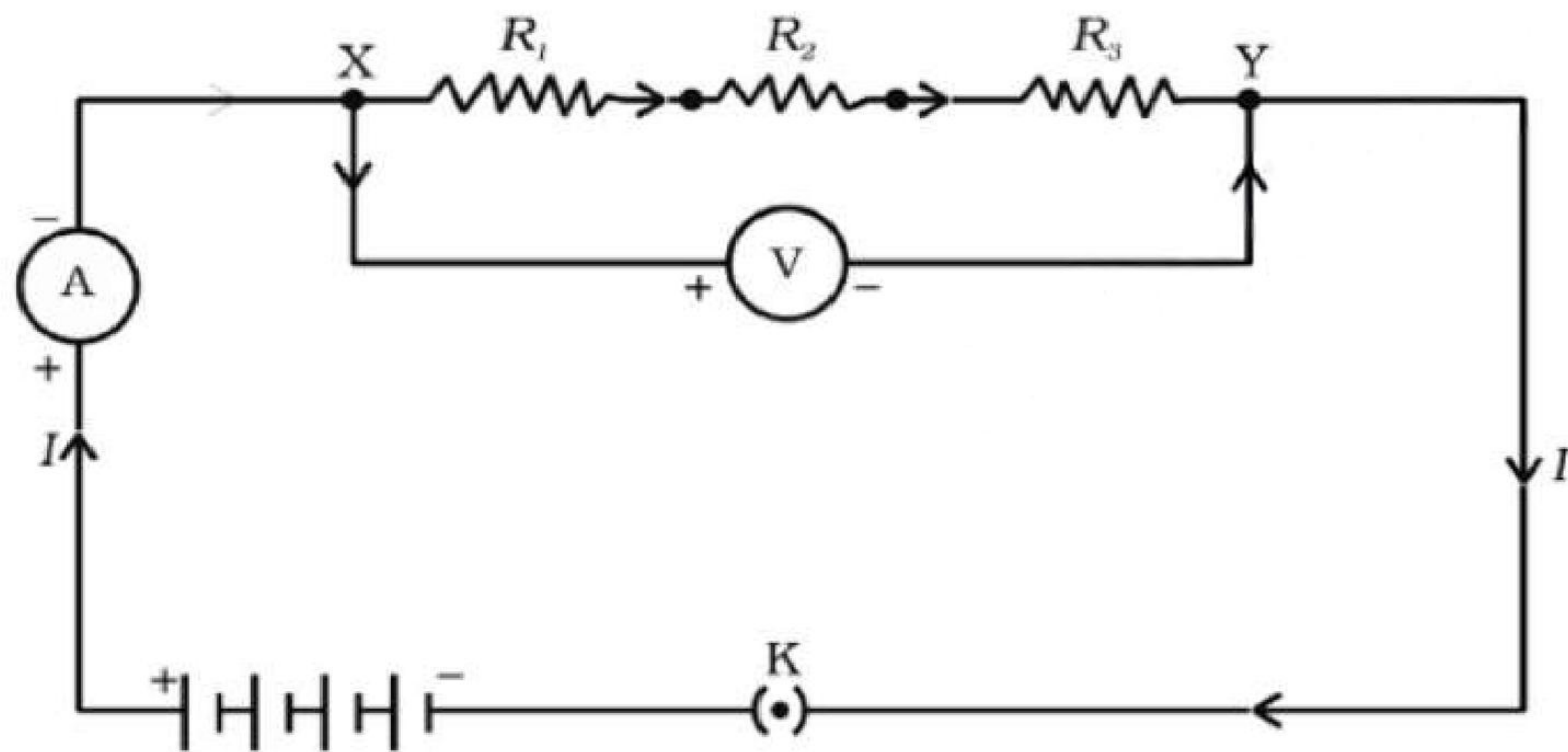




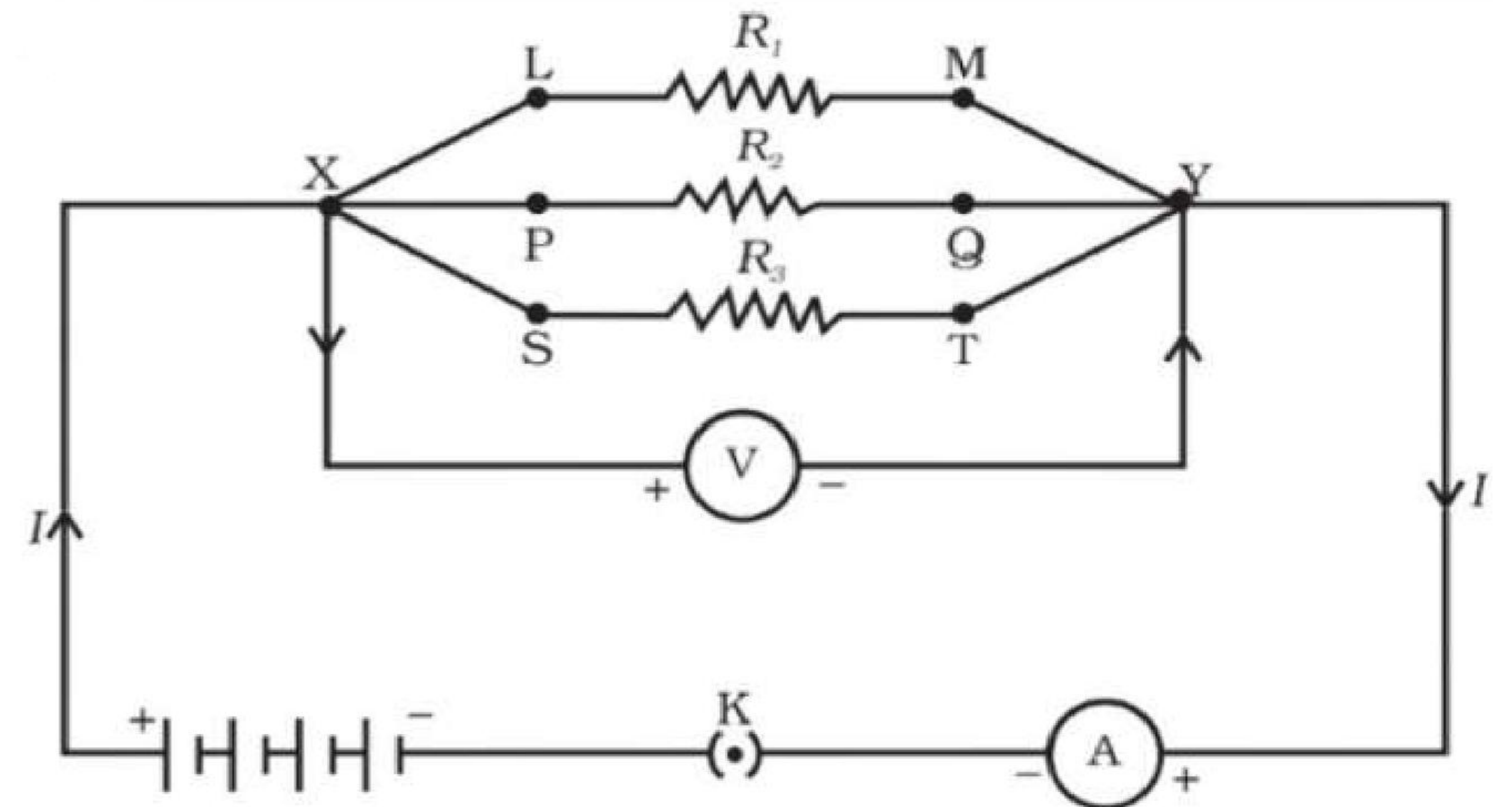
ELECTRICITY

RESISTANCE OF A SYSTEM OF RESISTORS

- In electrical gadgets, resistors are used in various combinations based on Ohm's law.
- There are 2 methods of joining the resistors: **Resistors in series** & **Resistors in parallel**.



Resistors in series

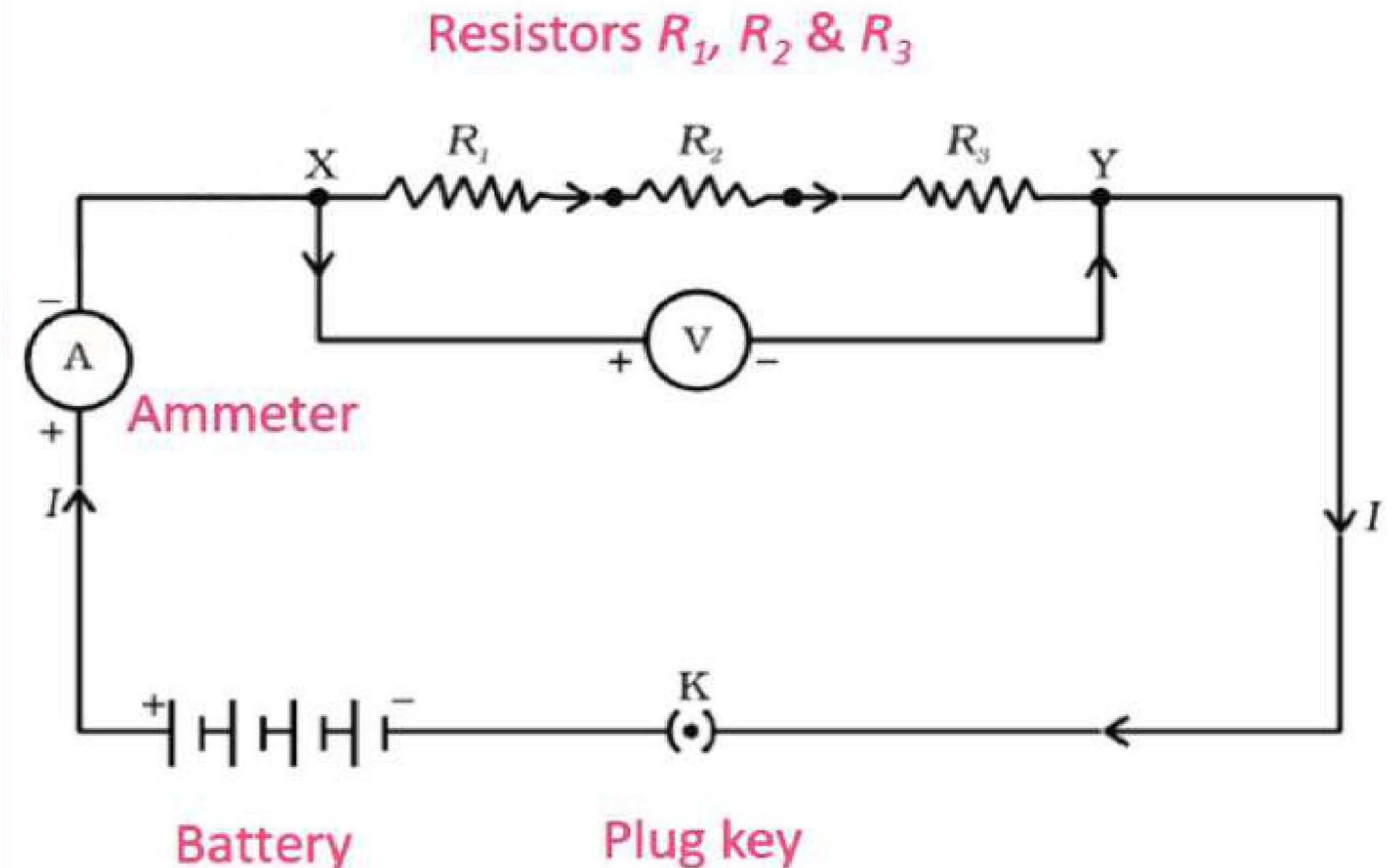


Resistors in parallel

RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Series

- Join three resistors having resistances R_1 , R_2 & R_3 (e.g. $1\ \Omega$, $2\ \Omega$, $3\ \Omega$) in series.
- Connect them with a 6 V battery, an **ammeter** and a plug key. Note the ammeter reading.
- Change the position of ammeter in between the resistors. The value is same.
- i.e., in a **series combination** of resistors, the **current is the same** in every part of the circuit or the same current through each resistor.

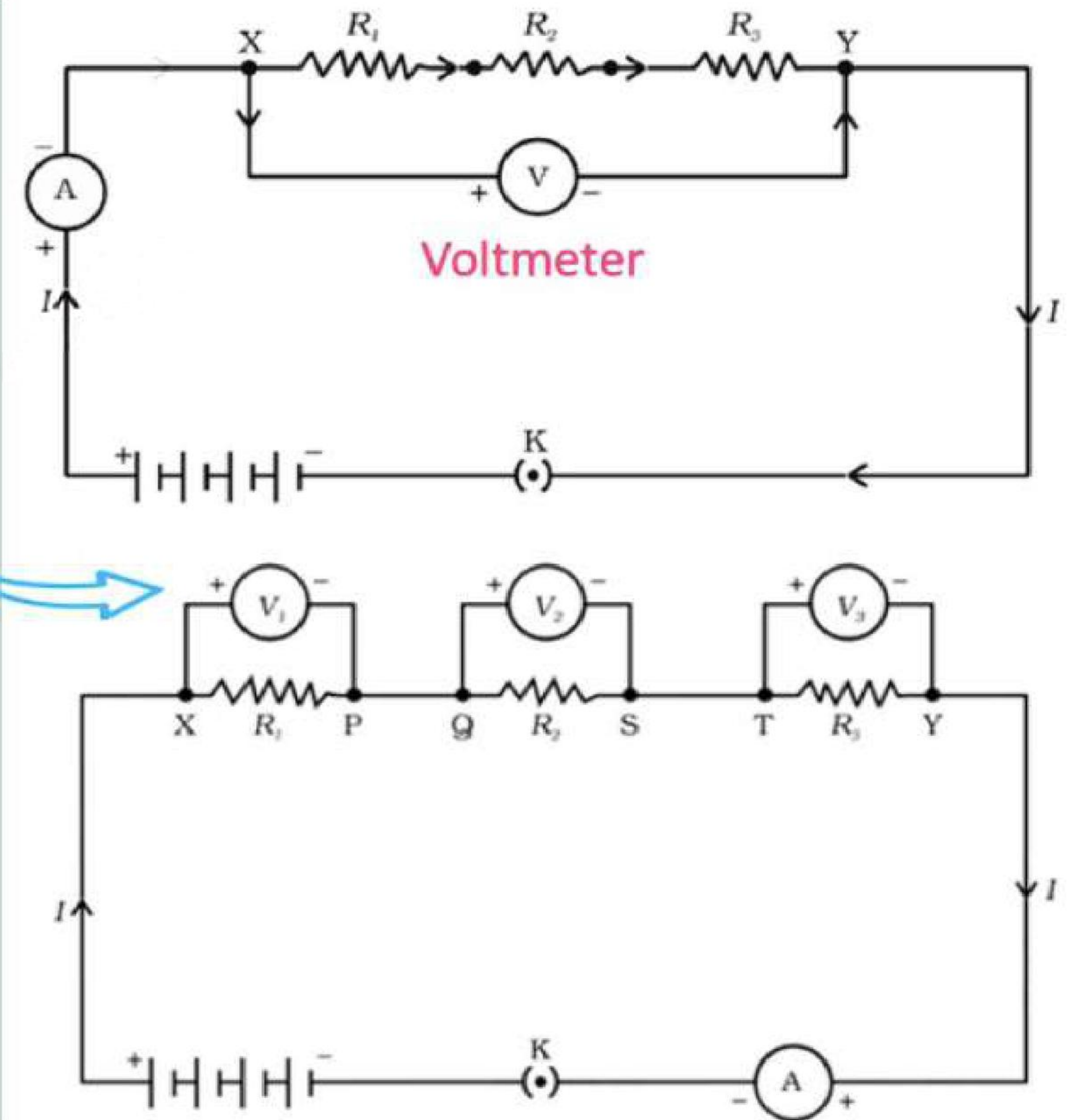


RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Series

- Insert a **voltmeter** across the ends X and Y of the series combination of three resistors. Note the **potential difference (V)**. Now measure the potential difference across the two terminals of the battery. Compare the two values.
- Now measure the potential differences V_1 , V_2 and V_3 across the first, second and third resistors separately.
- The total potential difference V across a combination of resistors in series is equal to the sum of potential differences across the individual resistors.

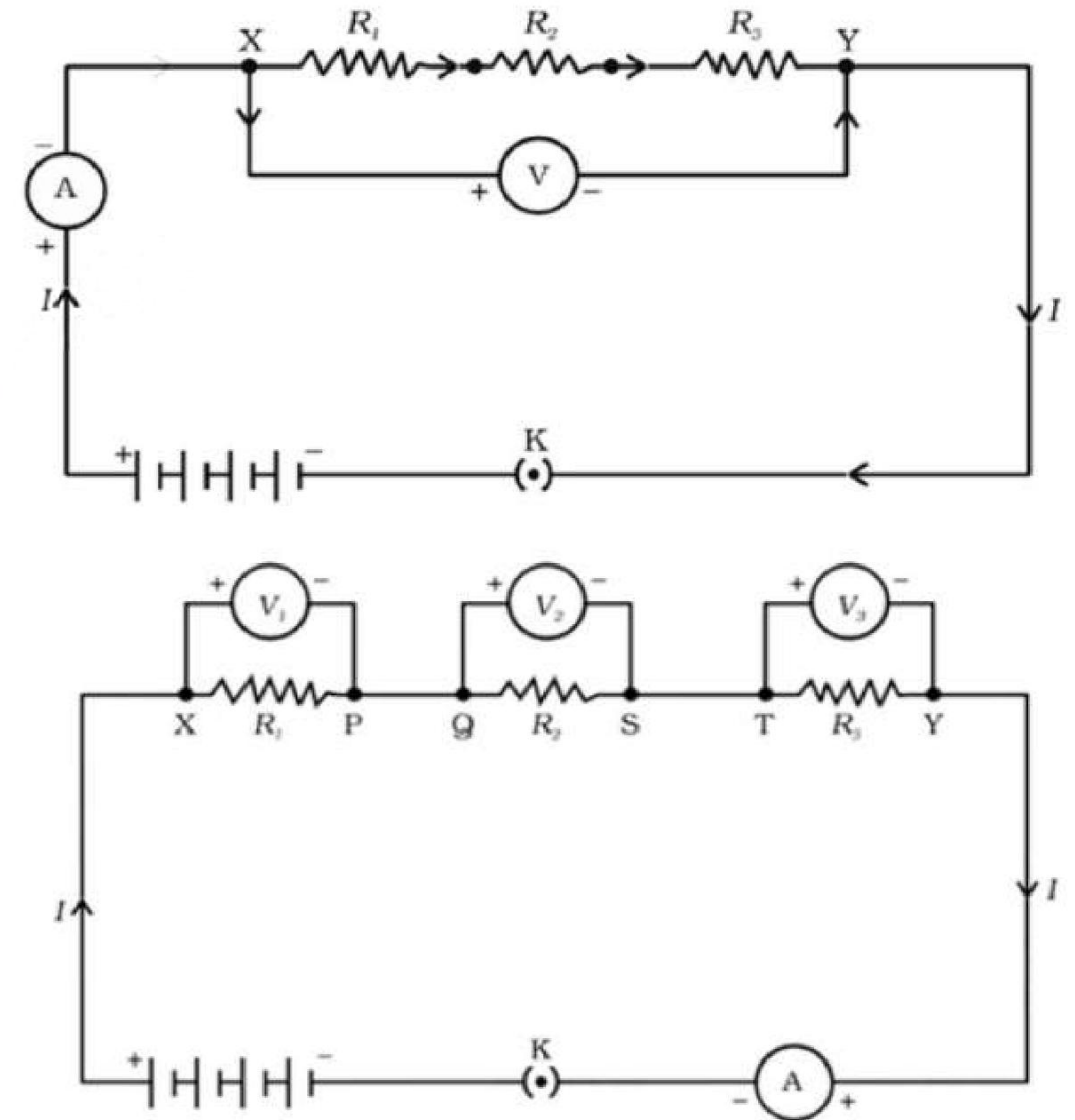
$$\text{i.e., } V = V_1 + V_2 + V_3$$



RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Series

- Let I be the **current** through this electric circuit. The current through each resistor is also I .
- The three resistors can be replaced by an equivalent single resistor of resistance R , such that the potential difference and the current remains the same.
- Applying the Ohm's law to the entire circuit, $V = I R$
- Applying Ohm's law to the three resistors separately,
$$V_1 = I R_1 \qquad V_2 = I R_2 \qquad V_3 = I R_3$$
$$I R = I R_1 + I R_2 + I R_3 \quad \text{or} \quad R_s = R_1 + R_2 + R_3$$
- When several resistors are joined in series, resistance of the combination R_s equals the sum of their individual resistances, R_1, R_2, R_3 .

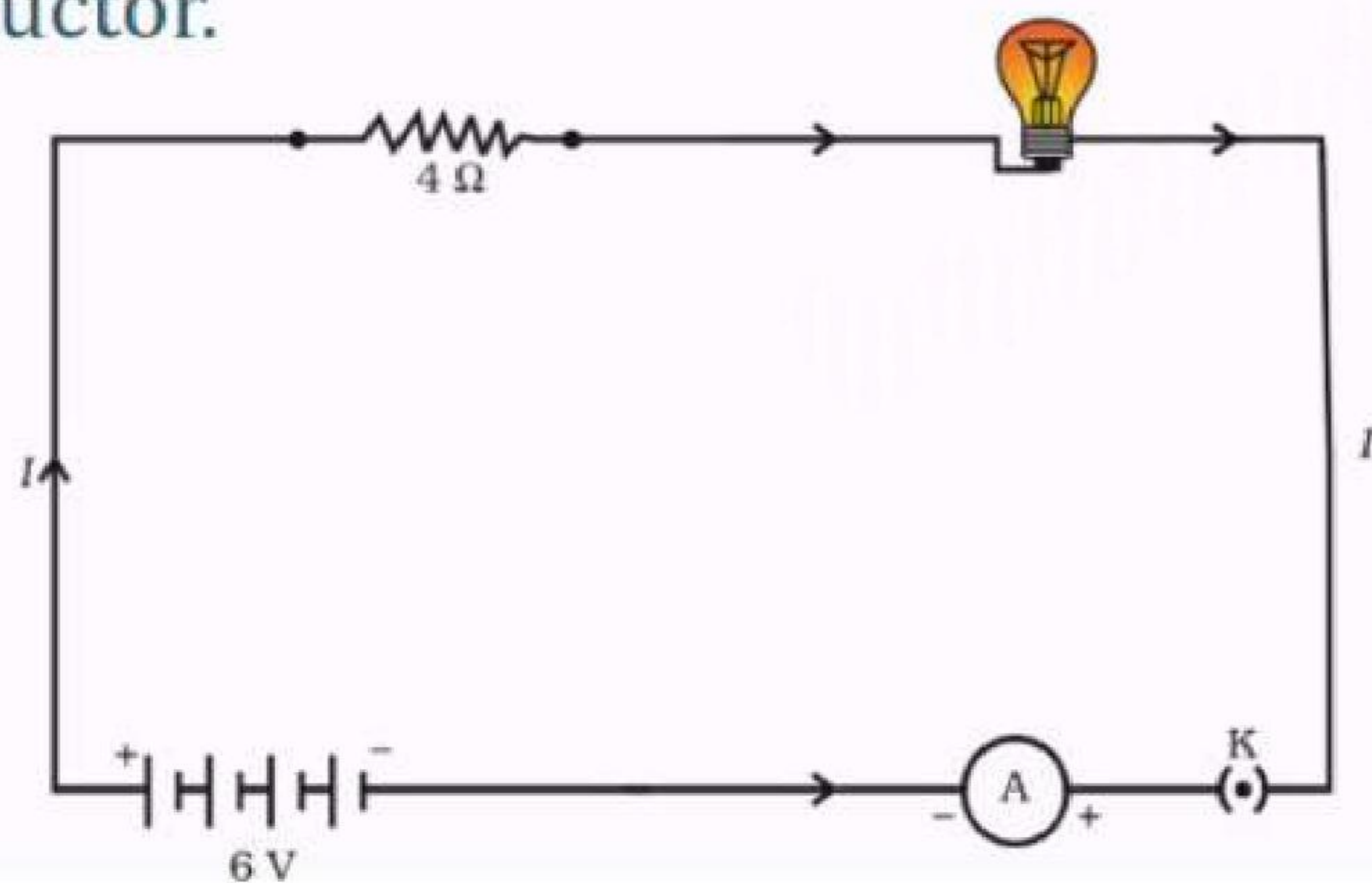


RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Series

Problem

An electric lamp, whose resistance is $20\ \Omega$, and a conductor of $4\ \Omega$ resistance are connected to a 6 V battery. Calculate (a) the total resistance of the circuit, (b) the current through the circuit, and (c) the potential difference across the electric lamp and conductor.



Solution

- a) Resistance of electric lamp, $R_1 = 20\ \Omega$
Resistance of the conductor, $R_2 = 4\ \Omega$.
 $\therefore$ Total resistance, $R_s = R_1 + R_2 = 20\ \Omega + 4\ \Omega = \underline{24\ \Omega}$.

- b) Total potential difference, $V = 6\text{ V}$.
Current through the circuit is $I = V/R_s$
 $= 6\text{ V}/24\ \Omega = \underline{0.25\text{ A}}$.

- c) Potential difference across the electric lamp:
 $V_1 = 20\ \Omega \times 0.25\text{ A} = \underline{5\text{ V}}$
Potential difference across the conductor:
 $V_2 = 4\ \Omega \times 0.25\text{ A} = \underline{1\text{ V}}$

If the series combination of electric lamp and conductor is replaced by an equivalent resistor, its resistance would be

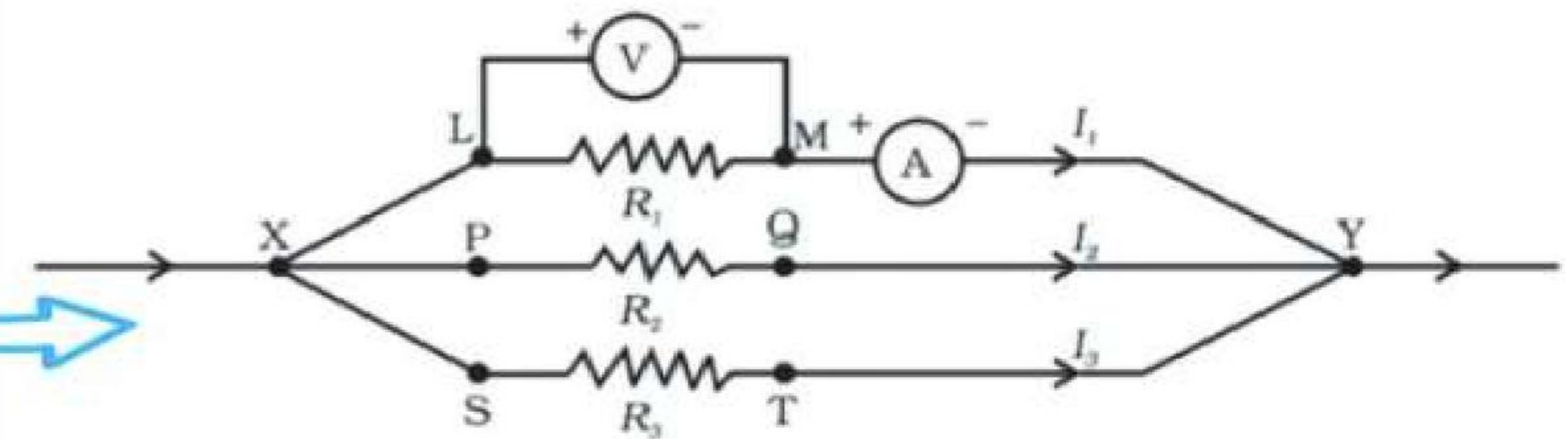
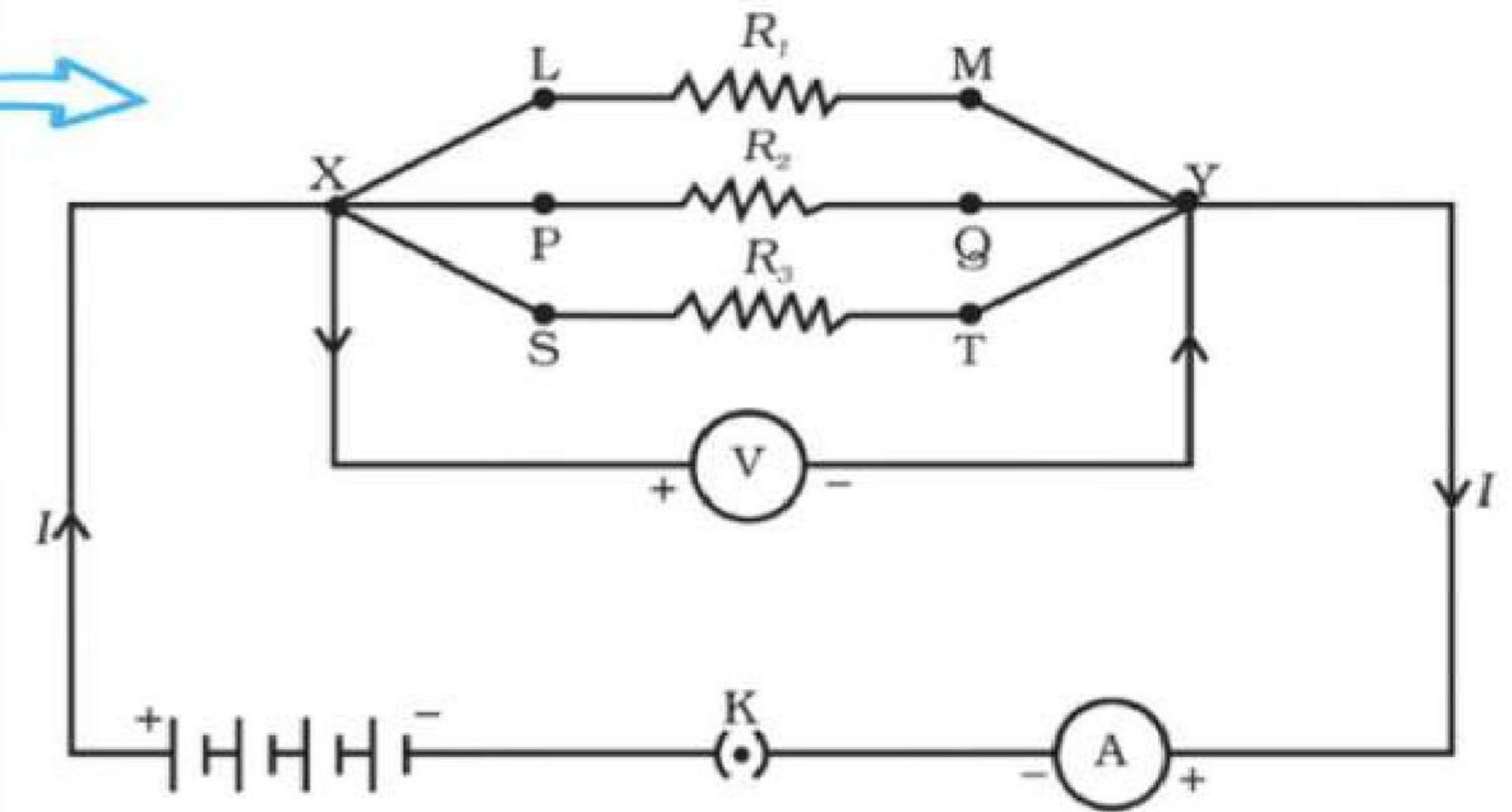
$$R = V/I = 6\text{ V}/0.25\text{ A} = \underline{24\ \Omega}$$

This is equal to the sum of the two resistances.

RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Parallel

- Make a **parallel combination, XY**, of three resistors having resistances R_1 , R_2 , & R_3 in an electric circuit. Connect a **voltmeter** in parallel with the resistors.
- Note the ammeter reading (I) and the voltmeter reading.
- Voltmeter shows the potential difference V , across the combination. The potential difference across each resistor is also V . This can be checked by connecting the voltmeter across each individual resistor.
- Insert the **ammeter** in series with the resistor R_1 . Note the ammeter reading, I_1 . Similarly, measure the currents I_2 & I_3 through R_2 & R_3 respectively.



RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Parallel

- The total current I , is equal to the sum of the separate currents through each branch.

$$I = I_1 + I_2 + I_3$$

- Let R_p be the equivalent resistance of the parallel combination of resistors.

$$\text{Hence, } I = V/R_p$$

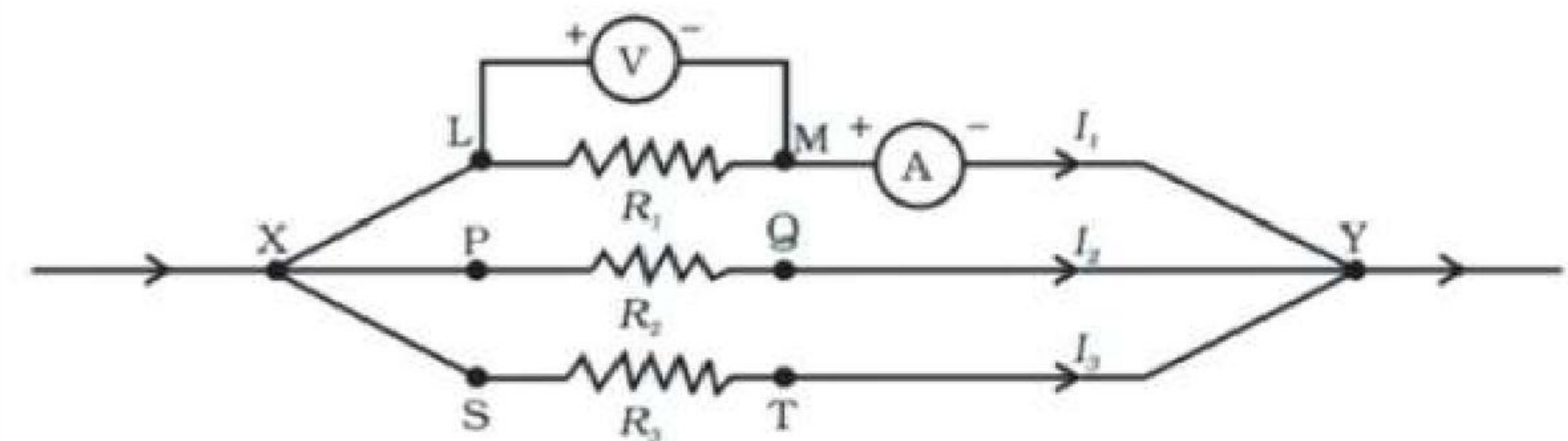
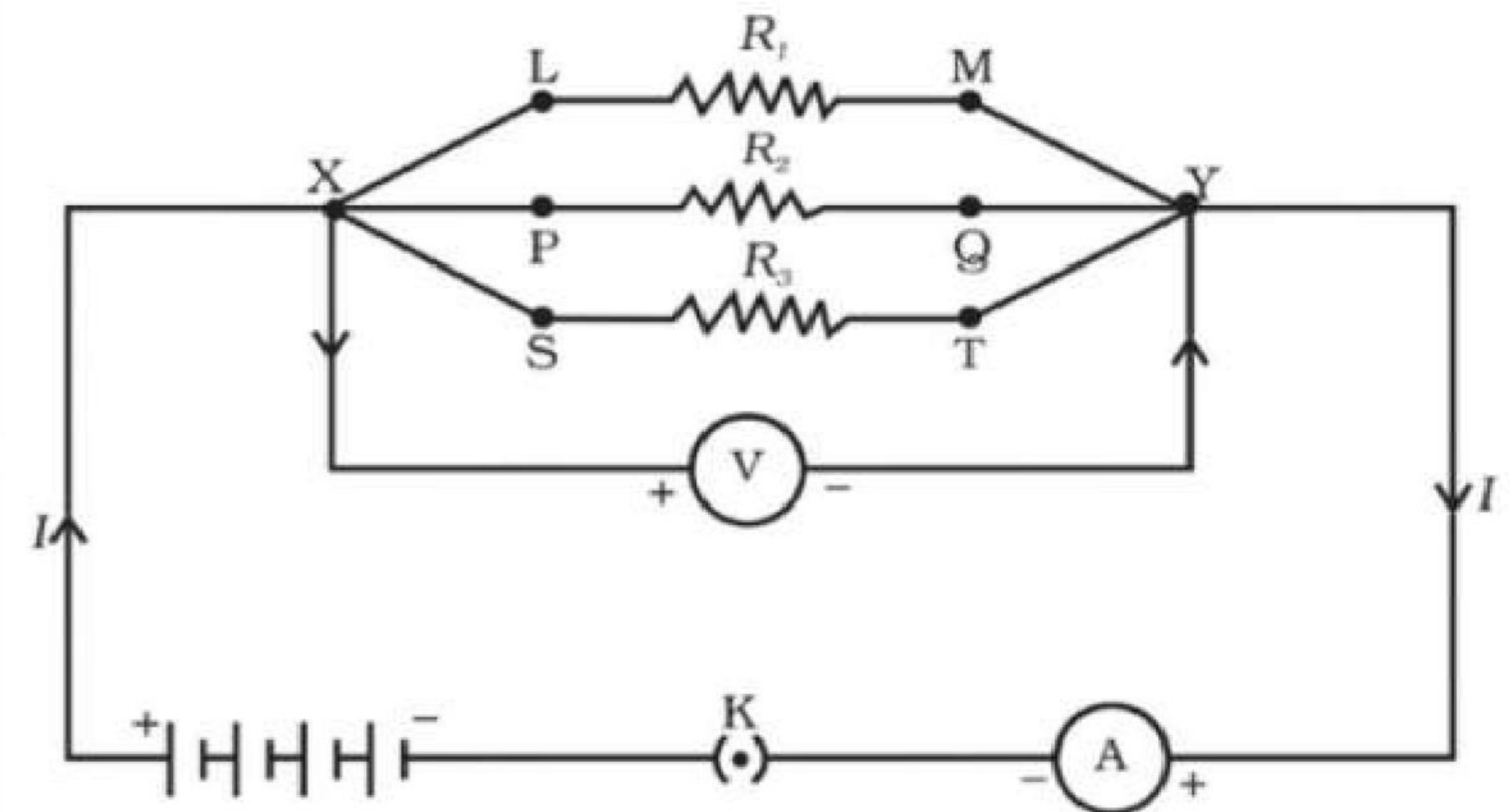
- Applying Ohm's law to each resistor,

$$I_1 = V/R_1 \quad I_2 = V/R_2 \quad I_3 = V/R_3$$

$$V/R_p = V/R_1 + V/R_2 + V/R_3 \quad \text{or}$$

$$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$$

- Thus, the reciprocal of the equivalent resistance of a group of resistances joined in parallel is equal to the sum of the reciprocals of the individual resistances.



RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Parallel

Problem

In the circuit diagram given, suppose the resistors R_1 , R_2 & R_3 have the values $5\ \Omega$, $10\ \Omega$, $30\ \Omega$, respectively, which have been connected to a battery of 12 V .

Calculate (a) the current through each resistor, (b) total current in the circuit, and (c) total circuit resistance.

Solution

$R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, and $R_3 = 30\ \Omega$.

Potential difference across the battery, $V = 12\text{ V}$.

This is also the potential difference across each of the individual resistor.

a) The current I_1 , through $R_1 = V/R_1 = 12\text{ V}/5\ \Omega = \underline{2.4\text{ A}}$

The current I_2 , through $R_2 = V/R_2 = 12\text{ V}/10\ \Omega = \underline{1.2\text{ A}}$

The current I_3 , through $R_3 = V/R_3 = 12\text{ V}/30\ \Omega = \underline{0.4\text{ A}}$

b) The total current in the circuit, $I = I_1 + I_2 + I_3$

$$= (2.4 + 1.2 + 0.4)\text{ A} = \underline{4\text{ A}}$$

c) The total resistance R_p , is $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$

$$1/R_p = 1/5 + 1/10 + 1/30 = 1/3$$

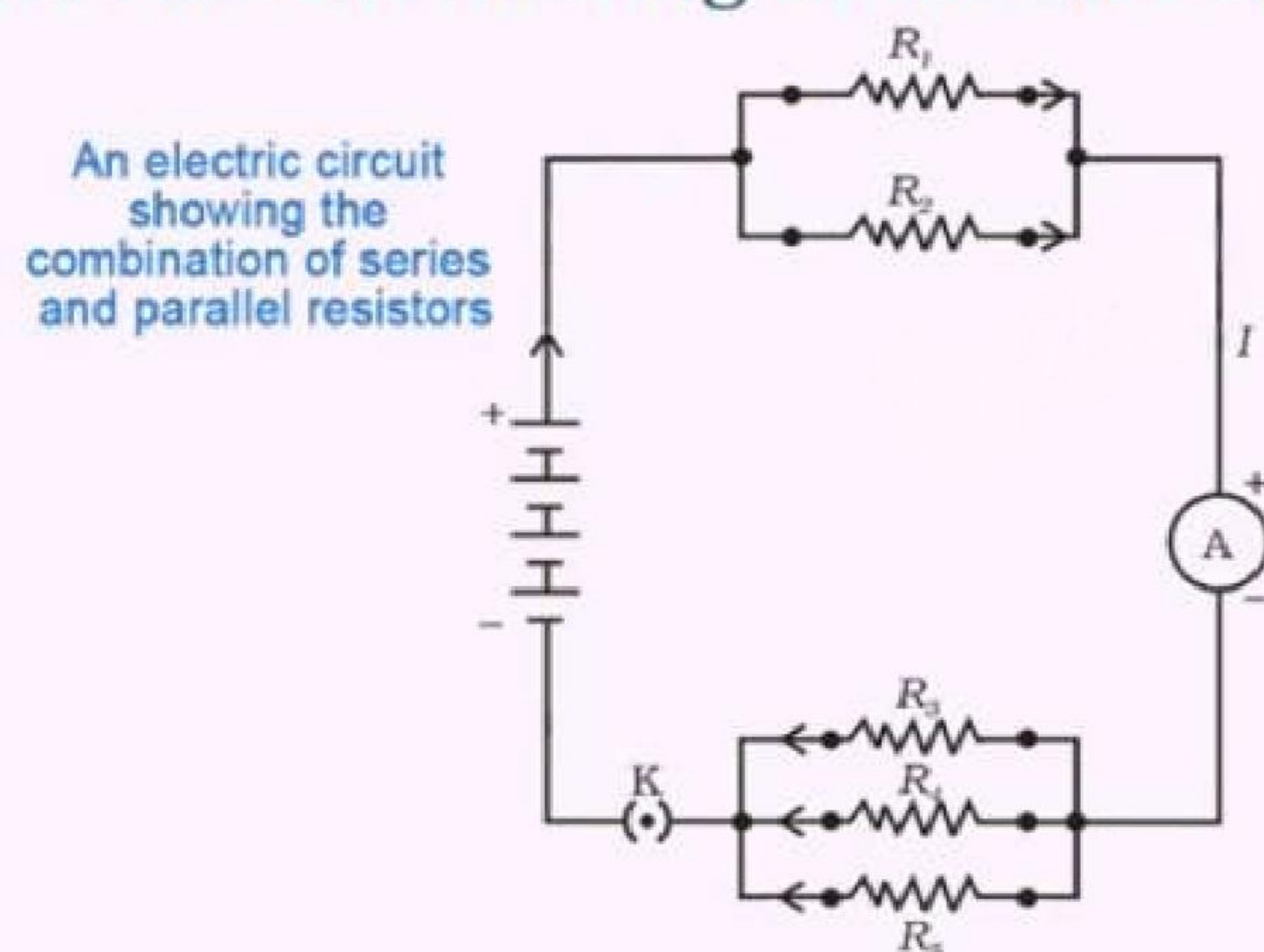
Thus, $R_p = \underline{3\ \Omega}$.

RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Parallel

Problem

If in Fig. given below, $R_1 = 10 \Omega$, $R_2 = 40 \Omega$, $R_3 = 30 \Omega$, $R_4 = 20 \Omega$, $R_5 = 60 \Omega$, and a 12 V battery is connected to the arrangement. Calculate (a) the total resistance in the circuit, and (b) the total current flowing in the circuit.



Solution

Suppose we replace the parallel resistors R_1 and R_2 by an equivalent resistor of resistance, R' . Similarly, we replace the parallel resistors R_3 , R_4 and R_5 by an equivalent single resistor of resistance R'' .

$$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$$

$$\therefore 1/R' = 1/10 + 1/40 = 5/40. \quad \text{i.e., } R' = \underline{\underline{8 \Omega}}$$

$$1/R'' = 1/30 + 1/20 + 1/60 = 6/60. \quad \text{i.e., } R'' = \underline{\underline{10 \Omega}}$$

$$\text{Thus, the total resistance, } R = R' + R'' = \underline{\underline{18 \Omega}}$$

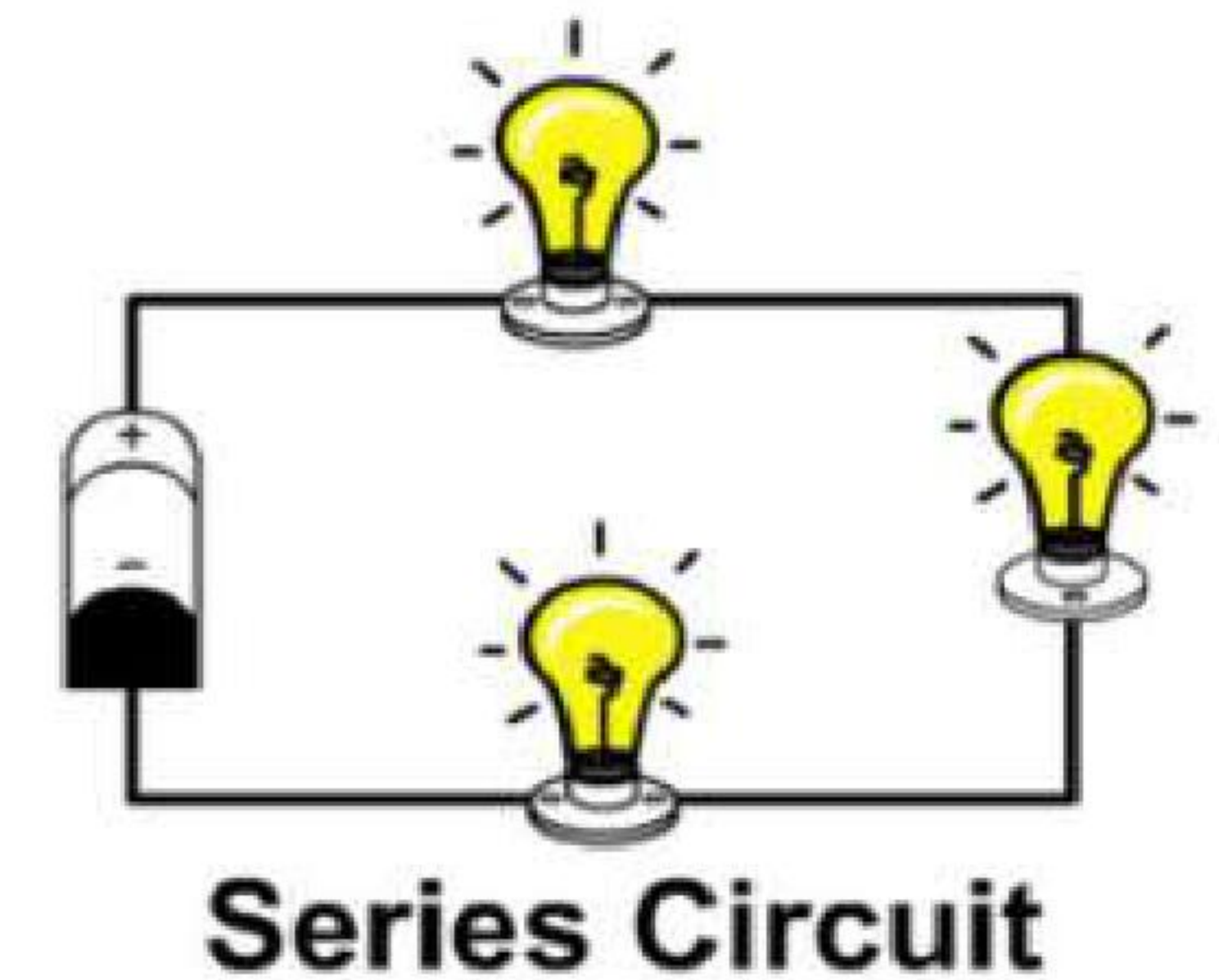
$$\text{The current in circuit, } I = V/R = 12 \text{ V}/18 \Omega = \underline{\underline{0.67 \text{ A}}}$$

RESISTANCE OF A SYSTEM OF RESISTORS

Resistors in Parallel

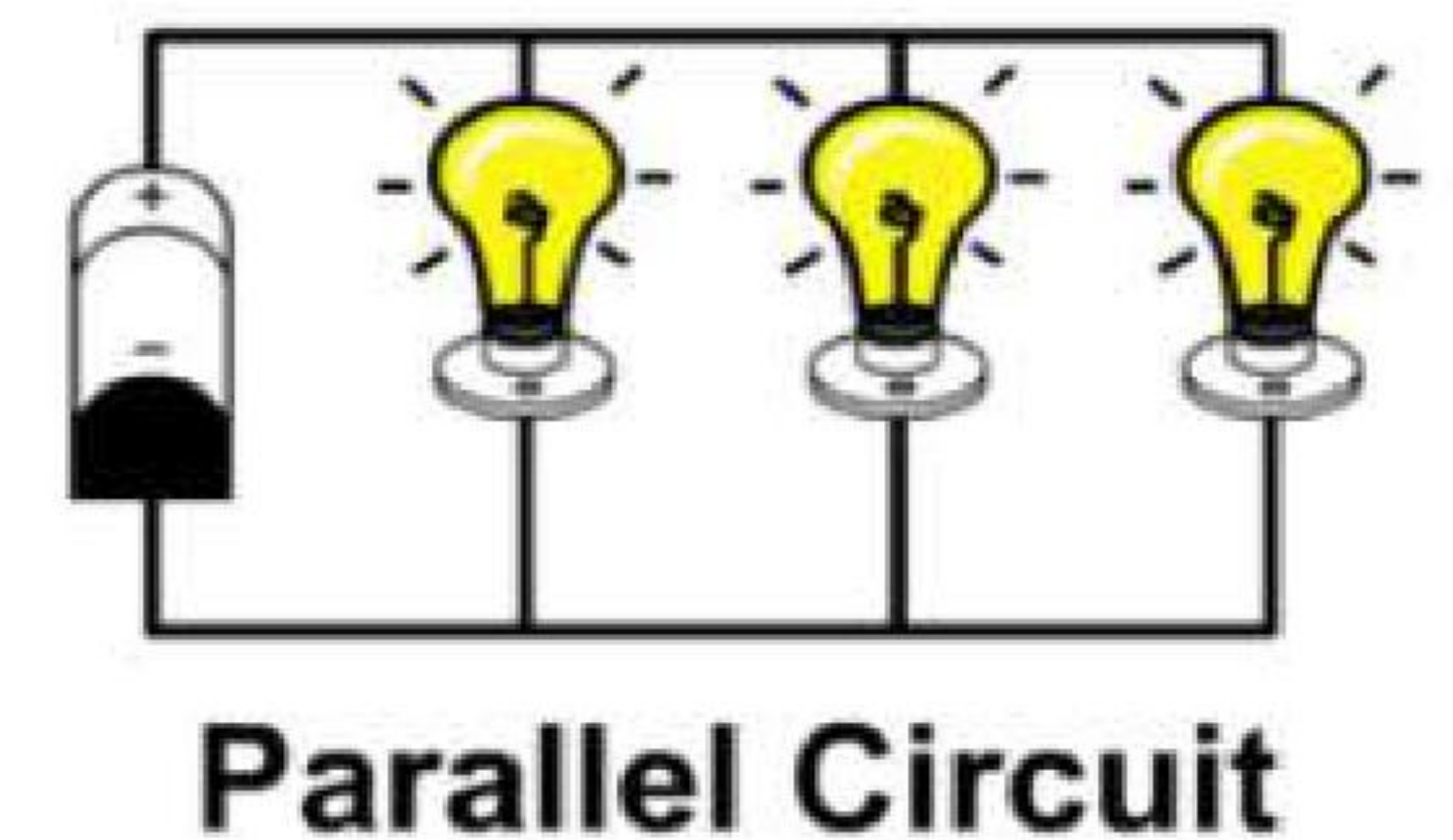
Disadvantages of Series Circuit

- ❖ The current is constant throughout the electric circuit. So, it is impracticable to connect an electric bulb and an electric heater in series, because they need currents of different values.
- ❖ When one component fails, the circuit is broken and none of the components works. E.g. it is very difficult to locate the dead bulb in fairy lights.



Advantages of Parallel Circuit

- ❖ It divides the current through the electrical gadgets.
- ❖ The total resistance is decreased. This is helpful when each gadget has different resistance and requires different current to operate properly.



HEATING EFFECT OF ELECTRIC CURRENT

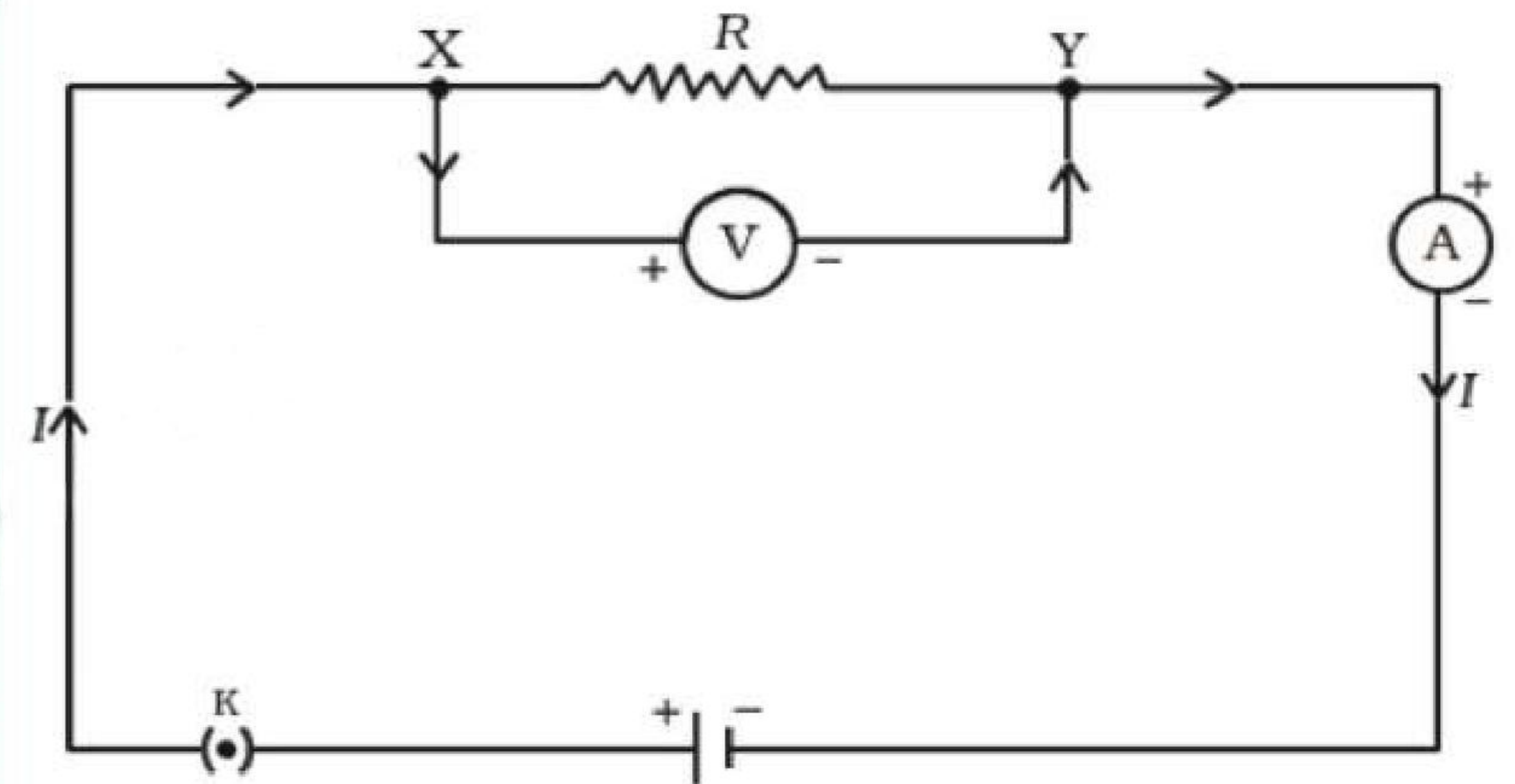
- A battery or a cell is a source of electrical energy. It generates potential difference that sets the electrons in motion to flow the current through a resistor or a system of resistors.
- A part of the source energy may be consumed into useful work (e.g. rotation of an electric fan). Rest of the energy is lost as **heat**. E.g., an electric fan becomes warm if used for longer time.



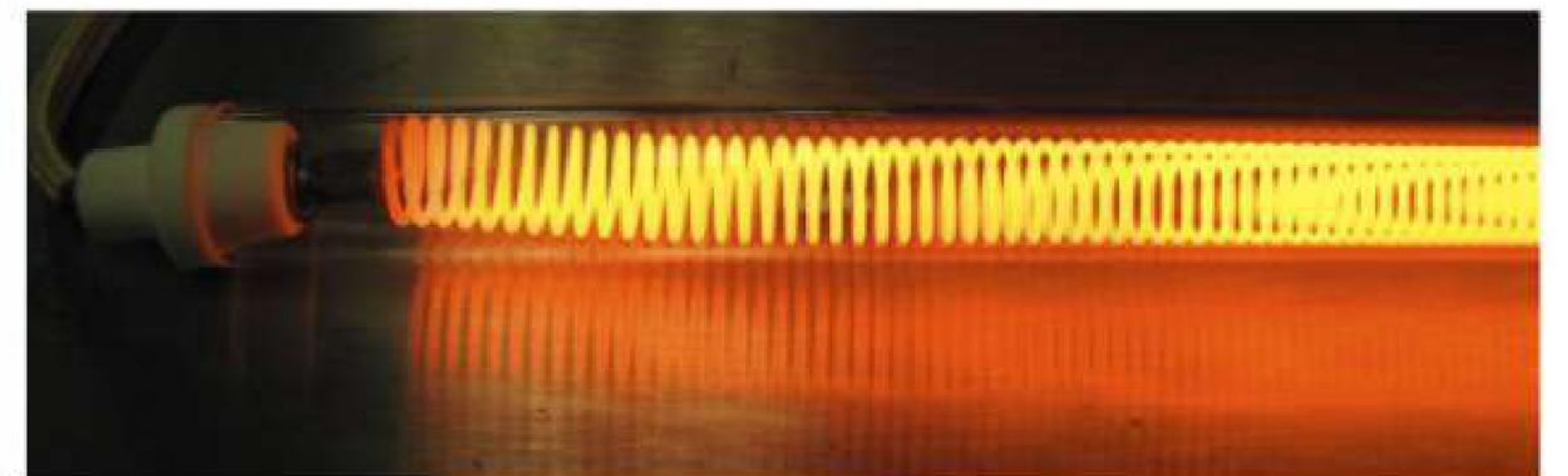
HEATING EFFECT OF ELECTRIC CURRENT

- If an electric circuit is purely resistive (i.e. a configuration of resistors only connected to a battery), the source energy is dissipated entirely as heat. This is called **heating effect of electric current**.
- Consider a **current I** flowing through a resistor of **resistance R** . Let the **potential difference** across it be **V** and **t** is the **time** during which a **charge Q** flows across.
- The work done in moving the charge Q through a potential difference V is **VQ** . Therefore, the source must supply energy equal to VQ in time t . Hence the **power** input to the circuit by the source is

$$P = V \frac{Q}{t} = VI$$



A steady current in a purely resistive electric circuit.

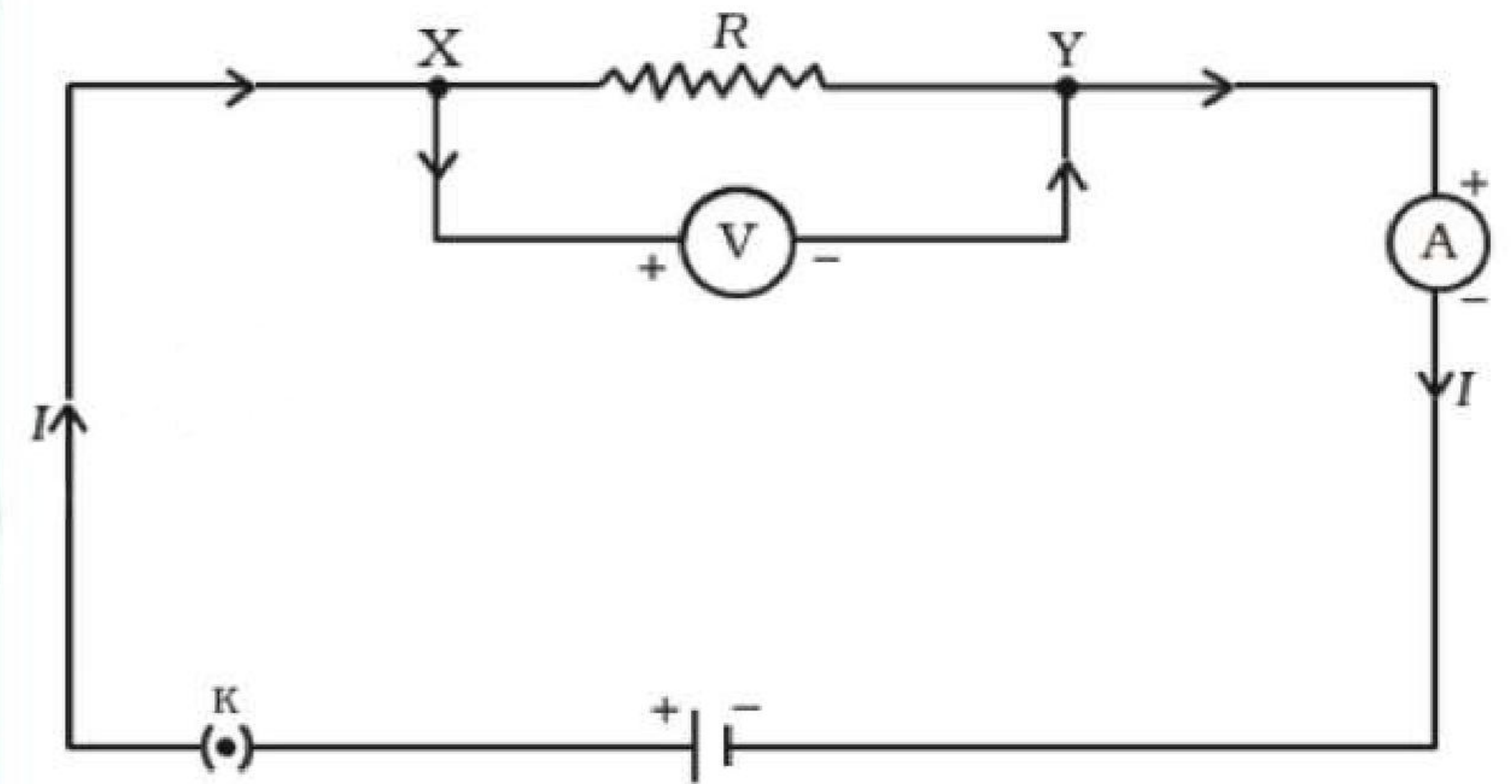


HEATING EFFECT OF ELECTRIC CURRENT

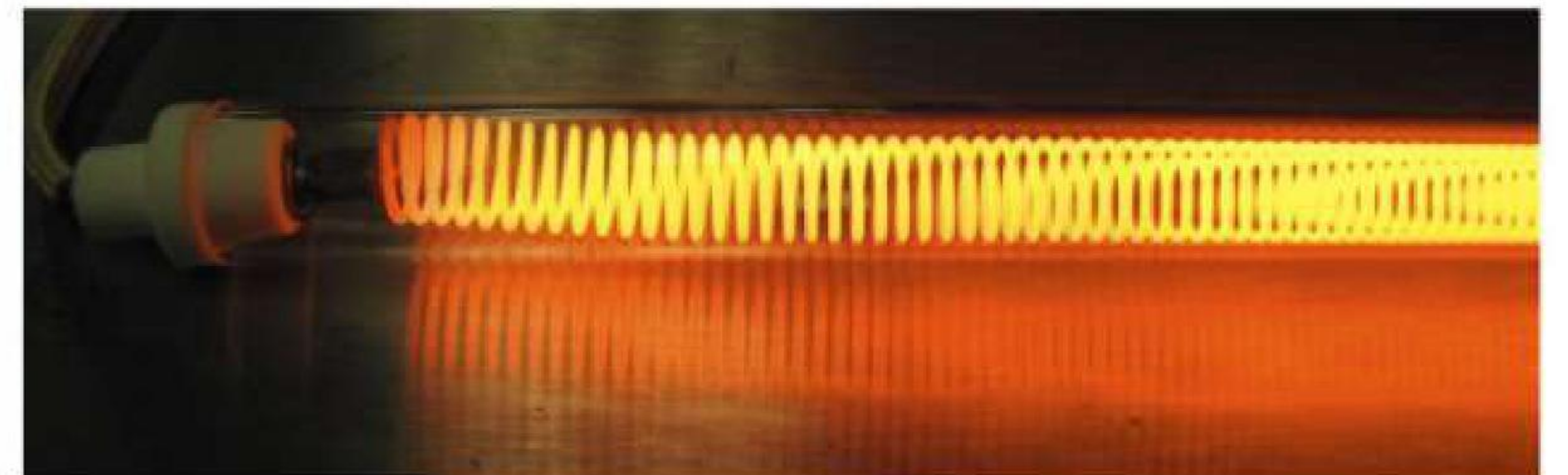
- Or the energy supplied to the circuit by the source in time t is $P \times t$, i.e., VIt . This energy is dissipated in the resistor as **heat (H)**.

Therefore, $H = VIt$

- Applying Ohm's law ($V = IR$), $H = I^2Rt$.
- This is called **Joule's law of heating**. It implies that heat produced in a resistor is directly proportional to
 - The **square of current** for a given resistance.
 - **Resistance** for a given current.
 - The **time** for which current flows through resistor.
- In practical situations, when an electric appliance is connected to a voltage source, Eq. $H = I^2Rt$ is used after calculating the current using the relation $I = V/R$.



A steady current in a purely resistive electric circuit.



HEATING EFFECT OF ELECTRIC CURRENT

Problem

An electric iron consumes energy at a rate of 840 W when heating is at the maximum rate and 360 W when the heating is at the minimum. The voltage is 220 V.

What are the current and the resistance in each case?

Solution

Power input, $P = VI$

Thus the current $I = P/V$

a) When heating is at the maximum rate:

$$I = 840 \text{ W} / 220 \text{ V} = \underline{3.82 \text{ A}}$$

Resistance of the electric iron is

$$R = V/I = 220 \text{ V} / 3.82 \text{ A} = \underline{57.60 \Omega}$$

b) When heating is at the minimum rate:

$$I = 360 \text{ W} / 220 \text{ V} = \underline{1.64 \text{ A}}$$

Resistance of the electric iron is

$$R = V/I = 220 \text{ V} / 1.64 \text{ A} = \underline{134.15 \Omega.}$$

HEATING EFFECT OF ELECTRIC CURRENT

Problem

100 J of heat is produced each second in a $4\ \Omega$ resistance. Find the potential difference across resistor.

Solution

$$H = 100\ \text{J}, \quad R = 4\ \Omega, \quad t = 1\ \text{s}, \quad V = ?$$

$$H = I^2 R t$$

So, the current through the resistor is

$$\begin{aligned} I &= \sqrt{(H/Rt)} \\ &= \sqrt{[100\ \text{J}]/(4\ \Omega \times 1\ \text{s})} \quad = \underline{5\ \text{A}} \end{aligned}$$

Thus the potential difference across the resistor is

$$V = IR = 5\ \text{A} \times 4\ \Omega = \underline{20\ \text{V}}$$

HEATING EFFECT OF ELECTRIC CURRENT

Practical Applications of Heating Effect of Electric Current

Due to heating effect, electrical energy is lost as heat. Also, it alters the properties of components in electric circuits. But **heating effect (Joule's heating)** has many applications:

- a. To make devices such as **electric laundry iron, electric toaster, electric oven, electric kettle & electric heater.**



Electric iron



Electric oven



Electric toaster



Electric kettle



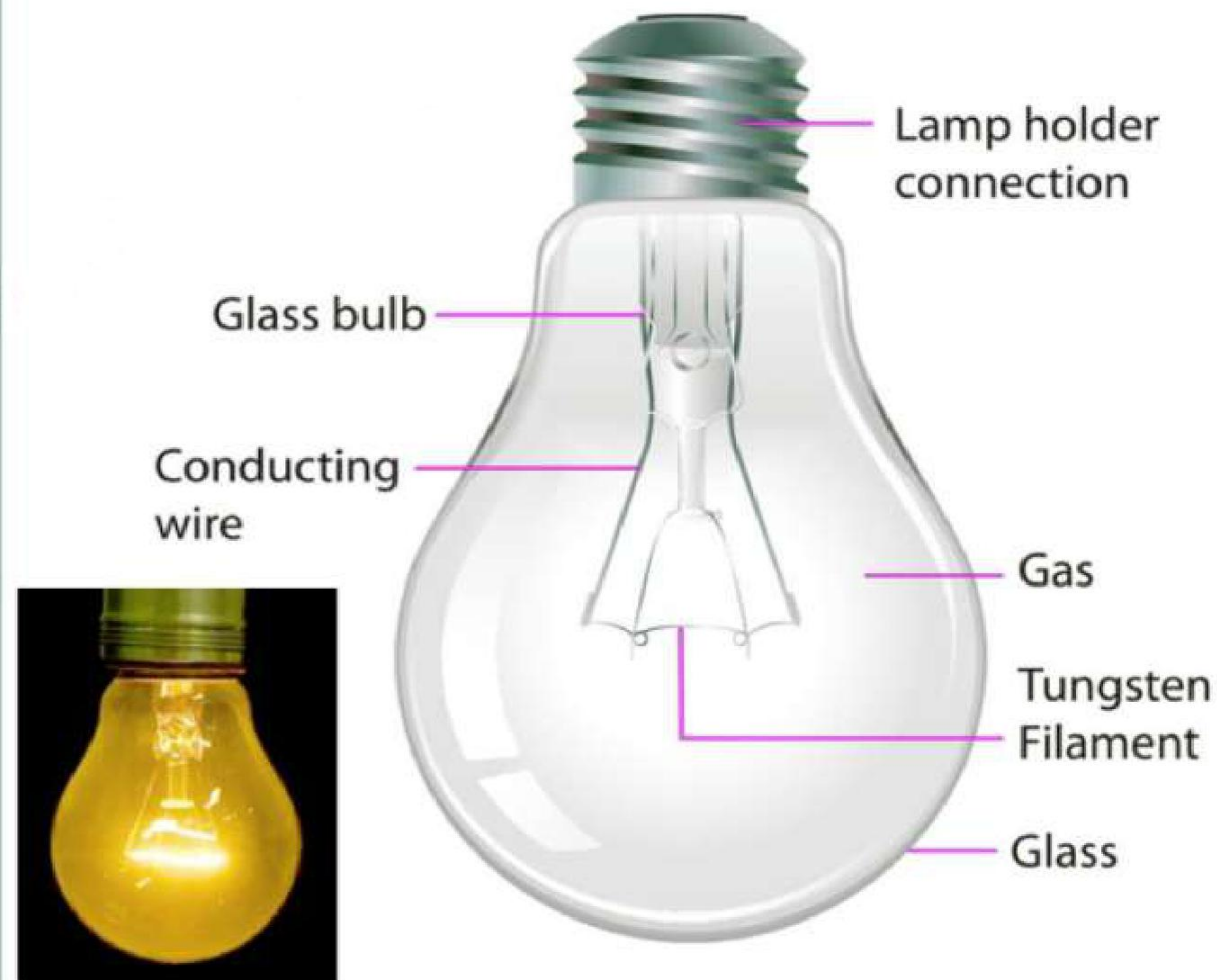
Electric heater

HEATING EFFECT OF ELECTRIC CURRENT

Practical Applications of Heating Effect of Electric Current

b. To produce **light in electric bulb**. Here, the **filament** made of metals with high melting point can retain much heat. So it gets very hot and emits light.

E.g. **tungsten** (melting point 3380°C) is used to make filaments. Filament should be thermally isolated, using insulating support. The bulbs are filled with chemically inactive **nitrogen** and **argon** gases to prolong the life of filament. Most of the power consumed by the filament appears as heat, but a small part is radiated as light.



HEATING EFFECT OF ELECTRIC CURRENT

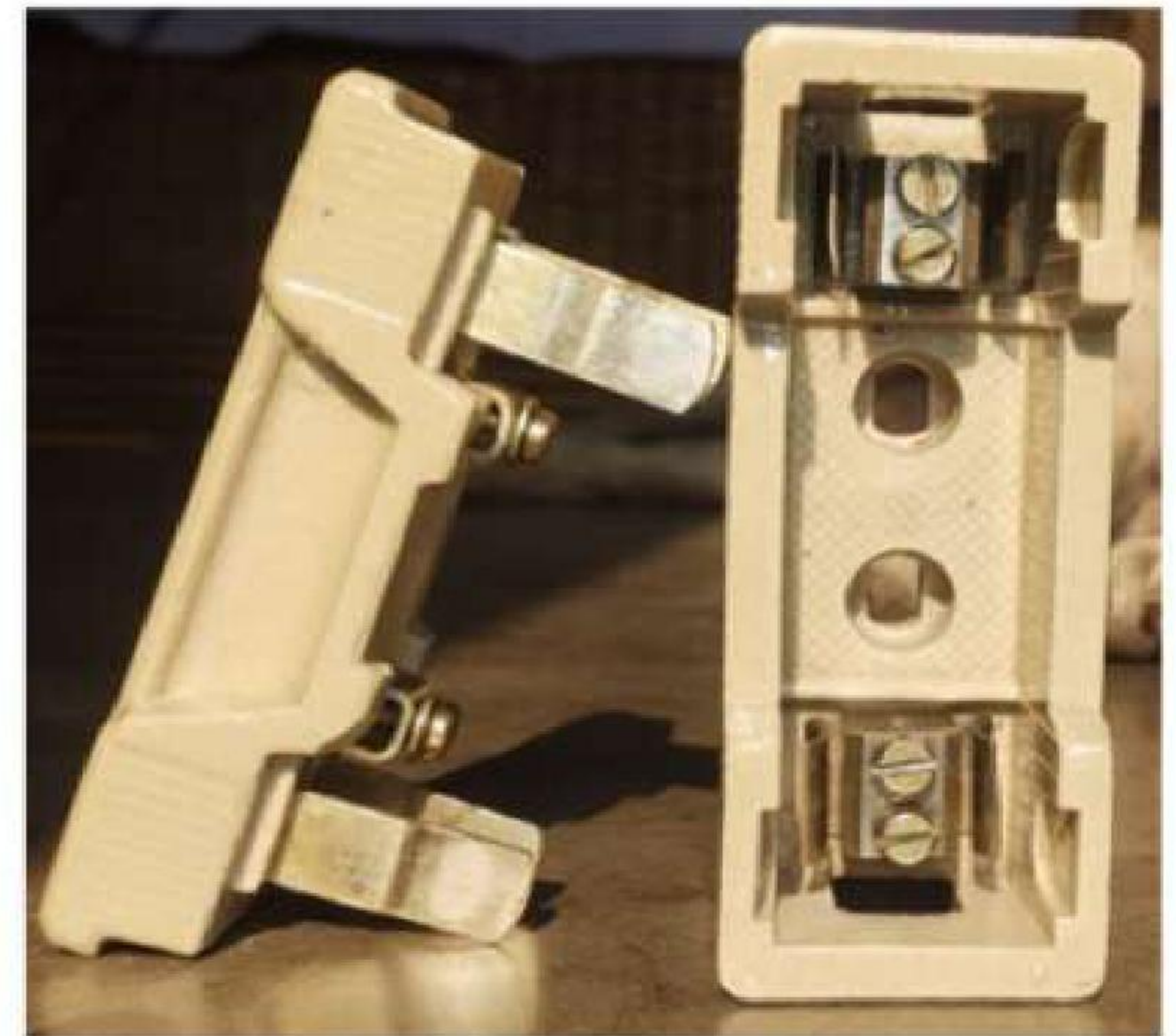
Practical Applications of Heating Effect of Electric Current

c. To make **fuse** used in electric circuits.

It protects circuits and appliances by **stopping the overflow of electric current**. The fuse is placed in series with the device. It consists of a **piece of wire** made of a metal or an alloy of suitable melting point (aluminium, copper, iron, lead etc.). During the overflow of the current, temperature of the fuse wire increases. It melts the fuse wire and breaks the circuit.

The fuse wire is encased in a **cartridge of porcelain** or similar material with metal ends.

The fuses used for domestic purposes are rated as **1A, 2A, 3A, 5A, 10A**, etc. E.g. when an electric iron which consumes 1 kW electric power is operated at 220 V, 4.54 A current ($1000/220$) flows in the circuit. In this case, a 5 A fuse must be used.



ELECTRIC POWER

- **Power** is the rate of doing work or rate of consumption of energy.
- Equation $H = I^2 R t$ gives the rate at which electric energy is dissipated or consumed in an electric circuit. This is also termed as **electric power**. The **power P** is given by

$$P = VI$$

Or $P = I^2 R = V^2 / R$

- The SI unit of electric power is **watt (W)**. It is the power consumed by a device that carries 1 A of current when operated at a potential difference of 1 V.

$$\text{Thus, } 1 \text{ W} = 1 \text{ volt} \times 1 \text{ ampere} = 1 \text{ V A}$$

- Watt is a very small unit. So, practically a much larger unit called **kilowatt** (1000 watts) is used.



ELECTRIC POWER

- Electrical energy is the product of power and time. Its unit is **watt hour (W h)**.
- One watt hour is the energy consumed when **1 watt of power** is used for **1 hour**.
- The commercial unit of electric energy is **kilowatt hour (kW h)**, commonly known as '**unit**'.
- $1 \text{ kW h} = 1000 \text{ watt} \times 3600 \text{ second}$
 $= 3.6 \times 10^6 \text{ watt second} = 3.6 \times 10^6 \text{ joule (J)}$
- In an electric circuit, electrons are not consumed. We pay for energy to move electrons through electric gadgets.



Energy meter

KSEB
Demand/Disconnection Notice
(Electricity Act 2003 P. 56)
Customer Care 1912
Navugal Electrical Section
0467-2203149

CH: 116701303

118 : 6701170904551
Name : VEDHAR V K
VEHARIL POODANKALL
le : T69.BCT N 11 / N8
ans : BESECOM
il Area : A05/4/164
ill Date : 07/09/2017
ue Date : 20/09/2017
sconn Dt : 16/10/2017
riff : LT-1R Don
urpose : Domestic
Deposit : 2000
uter(MN) Status OK
Load : 4 KW
Demand : 3.515 KVA
Phase : 1
rv Rd Dt : 07/07/2017
s Rd Dt : 07/09/2017
t Rd(DMF) : 1

Unit	Curr	Prev	Cons	Avg
M/R/I	4153	3790	363	208

Fixed Charges : 60.00
Meter Rent : 12.00
Energy Charges : 567.91
Duty : 48.84
Subsidies : -118.75

Bill Amount : 570.00
Adj : -93.22
Payable : 477.00

SHUN. P
Meter Reader

Pay On line: www.kseb.in
07-09-2017 10:06:22 AM

ELECTRIC POWER

Problem

An electric bulb is connected to a 220 V generator. The current is 0.50 A. What is the power of the bulb?

Solution

$$\begin{aligned}P &= VI \\&= 220 \text{ V} \times 0.50 \text{ A} \\&= 110 \text{ J/s} \\&= \underline{\underline{110 \text{ W}}}\end{aligned}$$

Problem

An electric refrigerator rated 400 W operates 8 hour/day. What is the cost of the energy to operate it for 30 days at Rs 3.00 per kW h?

Solution

$$\begin{aligned}\text{Total energy consumed by the refrigerator in 30 days:} \\400 \text{ W} \times 8.0 \text{ hr/day} \times 30 \text{ days} \\&= 96000 \text{ W h} &= \underline{\underline{96 \text{ kW h}}} \\ \therefore \text{cost of energy to operate the refrigerator for 30 days:} \\96 \text{ kW h} \times \text{Rs } 3.00 \text{ per kW h} &= \underline{\underline{\text{Rs } 288.00}}\end{aligned}$$